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ASSESSMENT OF *IN VIVO* MUTAGENIC POTENCY OF ETHYLENEDIAMINETETRAACETIC ACID IN ALBINO MICE

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Abstract—Ethylenediaminetetraacetic acid (EDTA) disodium salt, a widely used metal chelator, was studied for its potency to induce bone marrow micronuclei, dominant lethal mutations and sperm-head abnormalities in albino mice. The acute oral LD_{50} dose computed by probit regression was 30 mg/kg body weight in the strain used. Preliminary studies showed that oral administration of EDTA disodium salt at doses of 5, 10 and 15 mg/kg body weight/day on 5 consecutive days did not induce any obvious signs of toxicity. In the bone marrow micronucleus assay acute doses of EDTA disodium salt (5–20 mg/kg body weight) induced a dose-dependent increase in the incidence of micronucleated polychromatic erythrocytes at a 24-hr sampling. However, administration at doses of 5, 10 and 15 mg/kg for 5 consecutive days did not produce any observable effect on either the testicular or epididymal weights and histology. No appreciable alterations were observed in the caudal sperm counts at any of the sampling intervals and there was no treatment-related increase in the incidence of sperm-head abnormalities. Furthermore, treatment of male mice with EDTA disodium salt (10 mg/kg body weight/day for 5 consecutive days) induced no increase in the incidence of post-implantation embryonic deaths, except for a marginal but statistically insignificant increase during wk 2 and 3 of mating.

INTRODUCTION

Ethylenediaminetetraacetic acid (EDTA) is a widely used chelating agent in foods (Fishbein *et al.*, 1970; Rosival *et al.*, 1978) that also has innumerable applications in pharmacy and medicine (Fishbein *et al.*, 1970; *Food and Cosmetics Toxicology*, 1972). Because of its low toxicity (especially the Ca disodium salt), it is being allowed as a food additive in concentrations of up to 2.5 mg/kg (FAO/WHO, 1973; Furia, 1972). Because of the wide usage of EDTA and its salts in the human environment, the toxicological and mutagenic effects of these compounds have been thoroughly studied. EDTA is known to be excreted rapidly from the body (Foreman *et al.*, 1953) and its effects on serum enzymes, kidney, and respiratory and intestinal tracts are documented (Aronson and Ahrens, 1971; Braide and Aronson, 1974; Furia, 1972; Goyer *et al.*, 1978; Heller and Vostal, 1964).

The genetic toxicology of EDTA has been recently reviewed (Heindorff *et al.*, 1983). EDTA-induced inhibition of DNA synthesis was demonstrated in kidney cortex cells (Lieberman and Ove, 1962), in phytohaemagglutinin-stimulated lymphocytes (Alford, 1970) and in regenerating rat liver (Fujioka and Lieberman, 1964), and impairment of enzymes involved in DNA synthesis were implicated. EDTA has been shown to be non-mutagenic in bacterial systems such as *Escherichia coli* and *Salmonella typhimurium* (McCann *et al.*, 1975). In plant systems, EDTA has been shown to affect the chromosome morphology and mitosis (Biswas and Bhattacharyya, 1975; Davidson, 1958; Hyde, 1956) and the effect was speculated to be mediated by EDTA-induced cation deficiencies.

Data on the clastogenic potency of EDTA in animal systems are limited. Although it induced a low significant yield of chromosomal aberrations in the grasshopper (Saha, 1974), species-dependent differences were demonstrable (Ray-Chaudhuri, 1961). A positive mutagenic effect was also demonstrated in *Drosophila melanogaster* (Ondrej, 1965). Very few reports exist on the chromosome breaking ability of EDTA in mammalian systems. It was reported to induce chromosomal aberrations in bone marrow (Manna and Das, 1971) and splenic cells (Das and Manna, 1972) of mice. Similar observations were also reported in human leucocytes exposed to EDTA (Basrur and Baker, 1963). Furthermore, the role of EDTA in modulating the mutagenic response of a variety of physical and chemical mutagens has also been extensively studied (as reviewed by Heindorff *et al.*, 1983), although the molecular mechanisms leading to such potentiation are not clearly understood.

The wide usage of EDTA and its salts as food additives, its known clastogenic potency and its ability to potentiate the mutagenic response of chemical and physical agents clearly necessitates the generation of comprehensive data on its mutagenic potency in well validated *in vivo* mammalian systems. Accordingly, the primary objective of the present study was to assess the mutagenic potency of EDTA, using the bone marrow micronucleus assay and the dominant lethal

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Abbreviations: CP = cyclophosphamide; EDTA = ethylenediaminetetraacetic acid; EMS = ethyl methanesulphonate; MN = micronuclei; PCE = polychromatic erythrocytes.

test. Its potency to induce sperm-head abnormalities as well as its effect on epididymal sperm counts were also studied.

MATERIALS AND METHODS

Chemicals. The test chemical, EDTA disodium salt, and the positive mutagens, cyclophosphamide (CP) and ethyl methanesulphonate (EMS), were obtained from Sigma Chemical Co. (St Louis, MO, USA).

Animals and care. Adult Swiss albino mice of CFT strain (30–35 g) bred and reared by the Experimental Animal Production Facility at our research institute, were held in plastic cages in groups of six. During a week of acclimatization and throughout the experimental period, they were maintained on mouse pellets (Gold Mohur, Lipton India Ltd) *ad lib.* and had free access to water.

Study to determine oral LD_{50} dose. Groups of adult male mice (six per group) were intragastrically administered EDTA disodium salt dissolved in distilled water (1 mg/ml) at graded doses ranging from 5 to 50 mg/kg body weight. The LD_{50} value computed by probit regression analysis (Finney, 1977) was 30 mg/kg. However, the reported oral LD_{50} for EDTA in mice is 20.5 mg/kg body weight (Stecher, 1968).

Bone marrow micronucleus assay. 6-wk-old male mice (30 g) were intragastrically administered EDTA disodium salt (dissolved in distilled water) at doses of 5, 10, 15 and 20 mg/kg body weight. The animals were killed by cervical dislocation at 24 hr and bone marrow preparations were made using human AB serum as described previously (Muralidhara and Narasimhamurthy, 1990). The femoral cells were flushed out with human AB serum and the cellular suspension was spun at 2000 rpm for 5 min. Smears were prepared from the resulting pellet following the procedure of Schmid (1975). Four mice were used for each dose and four slides were prepared per animal and the incidence of micronuclei (MN) in an average of 4000 polychromatic erythrocytes (PCE) was scored per animal. Mice administered the carrier, distilled water (maximum 0.15 ml/mouse), or an acute dose of cyclophosphamide (20 mg/kg body weight ip) served as negative and positive controls, respectively. The MN data was analysed by Student's *t*-test.

Subacute treatment schedule: sperm morphology assay and caudal sperm counts. Groups of male mice (8–10-wk old) were administered (oral) distilled water (control) or EDTA dissolved in distilled water (1 mg/ml) at doses of 5, 10 and 15 mg/kg body weight (1/6, 1/3 and 1/2 LD_{50}) for 5 consecutive days. The animals were killed at 1, 3, 5 and 7 wk after the end of treatment. After recording the fresh weights of the epididymides and testes, the tissues were fixed in freshly prepared Bouin's fluid and processed for histological examination. Fresh cauda held in 0.9% NaCl were processed as described previously (Muralidhara and Narasimhamurthy, 1988) to determine the sperm count. Briefly, aliquots of sperm suspension were stained with 1% Eosin Y and the smears were scored for abnormal sperm-heads adopting the schedule of Wyrobeck and Bruce (1978). A minimum number of 1000 sperms were analysed per animal. Data on body weight, epididymal and testicular weights were analysed by Student's *t*-test. However, the data on sperm counts and sperm-head abnormalities for each week were analysed by the Mann–Whitney U-test.

Dominant lethal assay. Adult male mice (8 wk old) were administered (oral) 10 mg EDTA disodium salt/kg body weight/day in distilled water or water alone for 5 consecutive days. 15 males from each group were mated with virgin females (1:2 ratio) each week sequentially for a period of 8 wk. As a positive control, 15 male mice were administered (ip) an acute dose of 150 mg EMS/kg body weight and sequentially mated with adult virgin females (1:2 ratio) for 5 wk. The pregnant females were killed by cervical dislocation 14 days after the detection of vaginal plugs. Exposing the uterine horns, the contents were analysed for total implantation, early and late resorptions, and live and dead foetuses. Data on the average number of implantations, live embryos and the post-implantation embryonic deaths were compared on a 'per male basis' and statistically analysed by the Mann–Whitney U-test. Only males that led to pregnancies were considered, and each week of mating was considered separately.

RESULTS

Bone marrow micronucleus assay

The incidence of micronucleated PCE among the control mice was in the same range as reported previously by us (Muralidhara and Narasimhamurthy, 1990). The data shown in Table 1 clearly indicate that oral administration of EDTA induced a dose-dependent increase in the incidence of micronucleated PCE. However, the polychromatic erythrocytes/normochromatic erythrocytes ratio was not affected by EDTA treatment. The positive mutagen, CP, induced a highly significant increase in the incidence of micronucleated PCE ($P < 0.01$) clearly indicating the sensitivity of the assay strain. Although we are aware of the advantages of multiple dosing and multiple sampling protocols in such assays, we used an acute treatment schedule and a single sampling at 24 hr as this was a preliminary study. However, detailed studies are being conducted to determine the kinetics of micronuclei production by EDTA.

Subacute treatment

Oral administration of EDTA on 5 consecutive days at doses of 5, 10 and 15 mg/kg body weight did not induce any clinical signs of toxicity or mortality in mice. Neither the absolute nor relative weights of epididymides and testes were affected in the treatment groups (Table 2). On microscopic examination, no

Table 1. Incidence of micronuclei in bone marrow erythrocytes of male mice administered acute oral doses of EDTA

Dose mg/kg body weight	PCE with MN (%)	P/N ratio
Control (water)	0.350 ± 0.014	1.04
EDTA 5.0	0.543 ± 0.050†	1.08
10.0	0.650 ± 0.094†	0.98
15.0	0.850 ± 0.080*	0.97
20.0	1.433 ± 0.136*	0.95
CP 20.0	2.250 ± 0.166*	0.59

*$P < 0.01$.
†$P < 0.05$.
Values are means ± SEM of four animals (16,000 PCE) per dose, analysed by Student's *t*-test.

Table 2. Body weights, testicular and epididymal weights of male mice administered (oral) EDTA on 5 consecutive days

Wk	Organ weight (mg/100 g body weight)	Control	EDTA dose (mg/kg body weight) 5	10	15
1	Body weight (g)	46.0 ± 1.5	47.0 ± 0.5	47.0 ± 0.7	44.8 ± 1.2
	Testis	742 ± 44	716 ± 16	688 ± 22	727 ± 35
	Epididymis	77.3 ± 1.6	68.8 ± 3.4	73.1 ± 2.9	79.8 ± 2.9
3	Body weight (g)	44.3 ± 0.7	46.1 ± 0.9	44.1 ± 0.6	45.6 ± 1.3
	Testis	707 ± 32	681 ± 24	698 ± 46	607 ± 44
	Epididymis	74.5 ± 1.0	74.1 ± 2.8	72.1 ± 3.7	64.8 ± 3.3
5	Body weight (g)	47.5 ± 3.8	47.6 ± 1.2	50.1 ± 1.2	47.3 ± 1.1
	Testis	688 ± 35	613 ± 27	636 ± 7	642 ± 19
	Epididymis	73.6 ± 2.6	64.0 ± 2.7	60.8 ± 1.4	72.5 ± 0.7
7	Body weight (g)	46.1 ± 2.0	47.8 ± 3.0	46.0 ± 1.2	51.1 ± 1.4
	Testis	653 ± 42	621 ± 49	614 ± 20	563 ± 43
	Epididymis	74.2 ± 2.1	70.0 ± 5.9	75.8 ± 2.9	63.0 ± 2.4

Values are means ± SEM of six animals per dose group.
No significant differences between treatment and control groups (Student's *t*-test).

Table 3. Epididymal sperm counts and incidence of abnormal sperms in mice administered oral doses of EDTA on 5 consecutive days

Wk	Parameters	Control	EDTA dose (mg/kg body weight) 5	10	15
1	Sperm count*	117.8 ± 24.7	120 ± 15.4	123.0 ± 7.3	125.0 ± 8.9
	Abnormal sperms (%)	1.58 ± 0.40	1.78 ± 0.31	1.06 ± 0.09	1.63 ± 0.13
3	Sperm count	124.5 ± 12.5	119.4 ± 8.9	134.0 ± 12.4	123.0 ± 10.6
	Abnormal sperms (%)	1.84 ± 0.25	1.51 ± 0.15	1.48 ± 0.18	1.66 ± 0.17
5	Sperm count	128.8 ± 10.4	114.2 ± 15.1	118 ± 14.9	125.8 ± 12.5
	Abnormal sperms (%)	1.75 ± 0.35	1.48 ± 0.13	2.06 ± 0.21	1.20 ± 0.16
7	Sperm count	132.0 ± 13.6	126.0 ± 12.5	125.5 ± 10.5	128.5 ± 9.5
	Abnormal sperms (%)	1.65 ± 0.26	1.68 ± 0.11	0.91 ± 0.14	1.18 ± 0.15

*Values should be multiplied by 10^5 to give the actual sperm counts per cauda epididymis.
Values are means ± SEM for groups of six mice. No significant difference between treatment and control groups (Mann–Whitney U-test).

changes were evident in the histoarchitecture of epididymides and testes of mice dosed with EDTA.

Epididymal sperm counts and sperm morphology assay

The caudal sperm counts of EDTA-treated mice assayed at various intervals showed no significant alteration compared with controls at any of the tested doses (Table 3). Furthermore, there were no treatment-related changes in the incidence of sperm-head abnormalities, and the percentages of abnormal sperms in treated mice were on a par with those of controls (Table 3). The sperms analysed by us at 1, 3, 5 and 7 wk post-treatment presumably correspond with the treatment of mature sperms, spermatids, spermatocytes and spermatogonia, respectively.

Dominant lethal assay

The pregnancy frequency produced by EDTA-treated males was in the same range as that of controls except during the first week. While it ranged from 63.3 to 90% in treated males, it ranged from 83.3 to 93.3% among the controls. Furthermore, EDTA treatment had an observable effect on the mean implantation per litter at any of the sampling weeks (Table 4). However, a decrease was evident in the number of live embryos per litter during wk 2

Table 4. Results of dominant lethal assay in mice given 10mg EDTA/kg body weight for 5 consecutive days

Variable	Group	Wk 1	2	3	4	5	6	7	8
Total no. of litters	Control	25	25	26	28	27	28	28	28
	EDTA	19	22	23	27	23	26	25	23
	EMS	26	20	29	28	29	—	—	—
No. of implantations per litter	Control	10.7	10.6	11.5	10.7	11.7	11.8	11.0	11.9
	EDTA	10.1	10.3	10.8	10.6	10.3	11.5	11.5	11.9
	EMS	9.1†	6.5‡	9.7*	10.2	10.3	—	—	—
No. of live embryos per litter	Control	9.7	9.1	10.2	9.5	10.7	10.7	10.2	10.8
	EDTA	9.3	7.8	8.0	8.8	9.4	9.9	10.4	10.1
	EMS	2.9†	1.4‡	5.5†	9.0	9.3	—	—	—
No. of post-implantation deaths per litter§	Control	1.0 (9.3)	1.5 (14.2)	1.3 (11.3)	1.2 (11.2)	1.0 (8.5)	1.1 (9.3)	0.8 (7.3)	1.1 (9.2)
	EDTA	0.8 (7.9)	2.5 (24.3)	2.8 (25.9)	1.8 (17.0)	0.9 (8.7)	1.6 (13.9)	1.1 (9.6)	1.8 (15.1)
	EMS	6.2‡ (68.1)	5.1‡ (78.5)	4.2‡ (43.3)	1.2 (11.8)	1.0 (9.7)	—	—	—

Results are for groups of 15 males. The statistics were done using the male as the unit of treatment. Values marked with superscripts differ significantly (Mann–Whitney U-test) from the corresponding value (*$P < 0.02$; †$P < 0.01$; ‡$P < 0.002$).
§Values in parentheses are post-implantation deaths expressed as a percentage of the number of implants.

and 3. Consequently there was a marginal increase in the mean number of post-implantation deaths per litter during those weeks. In comparison, treatment of males with the positive mutagen, EMS, significantly affected all the three variables during the first 3 wk of mating, clearly indicating the sensitivity of mouse strain used (Muralidhara and Narasimhamurthy, 1988). Since EMS is known to induce dominant lethal mutations only during the post-meiotic stages of spermatogenesis, mating of EMS-treated males was restricted to 5 wk.

DISCUSSION

The available experimental evidence ascribes a low mutagenic activity and chromosome breaking ability to EDTA (Heindorff *et al.*, 1983) and this view is endorsed by the toxicological results obtained in laboratory animals. However, except for a few reports, there is no literature on the *in vivo* mutagenic activity of EDTA. Such data seem highly relevant considering the fact that this chelating agent is shown to modulate the genotoxic response of many environmental chemicals. Furthermore, no reliable data are available on the carcinogenicity of this compound (Heindorff *et al.*, 1983). In this context, our studies are aimed at assessing the mutagenic response of EDTA in both somatic and germ cells of mice.

The rodent bone marrow micronucleus assay is a very sensitive and reliable test that is being used as a screening test for *in vivo* chromosomal damage (Ashby *et al.*, 1988; MacGregor *et al.*, 1987; Schmid, 1976). The results of the present study clearly indicate the *in vivo* mutagenic activity of EDTA as it induced a dose-dependent increase in the incidence of micronucleated PCE, although it did not affect the P/N ratio. These results confirm the earlier observations of Manna and Das (1976), who reported a higher incidence of chromosomal aberrations in bone marrow cells of mice injected 0.05 M-EDTA.

EDTA administered at subacute doses failed to induce a significant mutagenic response in germ cells of mice. It did not affect significantly either epididymal or testicular weights. The non-interference of EDTA in sperm production and its development is suggested by the normal caudal sperm counts and normal frequency of sperm-head abnormalities observed in EDTA administered to mice at all dosages. Previously, many authors have reported the ability of EDTA to induce dominant lethal mutations in insect species such as wasps (LaChance, 1959), and *D. melanogaster* (Baranauskajte *et al.*, 1972; Ondrej, 1965). However, our results clearly show that it failed to increase post-implantation embryonic deaths, although we did observe a marginal decrease in the number of live embryos during wk 2 and 3.

Thus, the present study clearly showed that EDTA administered orally at acute doses induced a significant mutagenic response in bone marrow erythrocytes but was ineffective in inducing either sperm-head abnormalities or dominant lethal mutations at the tested subacute doses.

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